

Petr ŠIMČÁK



ABOUT ME

Former theatre and film actor who transitioned into software development through technical education and project-based work. I now focus on practical problem solving in C, C++, and Python, while bringing communication skills, discipline, adaptability, and experience working in demanding team environments.

CONTACT

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TECHNICAL SKILLS

- C, C++, Python, Bash, MATLAB, Typescript, SQL
- Git, Docker, Linux, Nginx, LaTeX
- Systems programming, networking
- Backend development, signal processing

LANGUAGES

- Czech (C2)
- English (C1)
- German (B1)

HIGHLIGHTS

- 2nd place at Škoda Hackathon (team lead / backend role)
- Hitchhiker at 42Prague
- 3D printer assembly, maintenance, and support
- C++ exam board support

SELECTED PROJECTS

2026

FT_TRANSCENDENCE (*full-stack web application*)

- ◇ Team-based web application developed at 42. Audiobook marketplace with audio streaming, built with FastAPI, React, Docker, and PostgreSQL.

2026

FT_IRC (*C++98, sockets, poll*)

- ◇ Built a minimal multi-client IRC server in a team of three; worked mainly on server-side structure and IRC commands.

2025

CUB3D (*C, MLX42*)

- ◇ Simplified Wolfenstein-style raycasting engine; implemented parsing and core raycasting logic.

2025

BACHELOR THESIS: HEART RATE ESTIMATION FROM PPG SIGNALS (*Python, signal processing*)

- ◇ Compared multiple methods for heart-rate estimation from physiological signals and worked on automatic signal-quality assessment.

2023–2025

ADDITIONAL 42 PROJECTS (*C, C++, Docker*)

- ◇ Completed minishell, Inception and other projects; gained experience with Unix processes, shell logic, containerization, and deployment setup.

EDUCATION

2023–2026

42PRAGUE (*with period at 42Heilbronn, 2024–2025*)

- ◇ Software engineering curriculum based on project work, peer learning, and practical problem solving.

2020–2025

BRNO UNIVERSITY OF TECHNOLOGY

- ◇ Bachelor's Degree in Sport Technology. Thesis: *Heart Rate Estimation from the PPG Signals*.

ADDITIONAL EXPERIENCE

2026–present

HITCHHIKER AT 42PRAGUE

- ◇ Representing the school values and culture, supporting new students, and helping with school events.

2025

ŠKODA B3R BRIGADE — *Team Lead / Backend Developer*

- ◇ 2nd place at Škoda Hackathon for a talent-matching application built with FastAPI, React, Docker, and SQLite.

2022–present

3D PRINTER MAINTENANCE AND SUPPORT

- ◇ Assembled a 3D printer at BUT and later maintained and supported its operation at 42Prague.

2013–2021

MOVIE ACTOR (*IMDb*)

- ◇ Worked in long-term collaborative productions in demanding environments.

2009–2015

THEATRE ACTOR (*Brno City Theatre*)

- ◇ Professional stage experience in ensemble-based productions.